

$$\bar{A} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \quad \bar{B} = \begin{bmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \end{bmatrix} \quad \begin{aligned} u_1 &= b_{11}V_1 + b_{12}V_2 + b_{13}V_3 \\ u_2 &= b_{21}V_1 + b_{22}V_2 + b_{23}V_3 \end{aligned}$$

$$a_{11} = - \frac{(1-D_1)R_{B1}(\Sigma R - R_{B1}) + D_1 R_{B2}(\Sigma R - R_{B2}) + \Sigma R(R_S + R_L)}{L_1 \Sigma R}$$

$$a_{12} = \frac{R_{B2}[D_2 R_{B3} + (D_1 - D_2)(\Sigma R - R_{B2}) + (1-D_1)R_{B1}]}{L_1 \Sigma R}$$

$$a_{21} = \frac{D_1 R_{B2} R_{B3} - (D_2 - D_1) R_{B1} R_{B3} + (1-D_2) R_{B1} R_{B2}}{L_2 \Sigma R}$$

$$a_{22} = - \frac{(1-D_2)R_{B2}(\Sigma R - R_{B2}) + D_2 R_{B3}(\Sigma R - R_{B3}) + \Sigma R(R_S + R_L)}{L_2 \Sigma R}$$

$$D_2(t) = D_2 + \hat{d}_2$$

$$\begin{bmatrix} \hat{i}_{L1} \\ \hat{i}_{L2} \end{bmatrix} = \begin{bmatrix} \hat{i}'_{L1} \\ \hat{i}'_{L2} \end{bmatrix} + \begin{bmatrix} \hat{i}_{L1} \\ \hat{i}_{L2} \end{bmatrix}$$

$$\begin{bmatrix} \dot{\hat{i}}_{L1} \\ \dot{\hat{i}}_{L2} \end{bmatrix} = \bar{A} \begin{bmatrix} \hat{i}_{L1} \\ \hat{i}_{L2} \end{bmatrix} + (\bar{A} \begin{bmatrix} \hat{i}'_{L1} \\ \hat{i}'_{L2} \end{bmatrix} + \bar{B}u) \hat{d}_2$$

$$\left[\begin{aligned} m &= R_{B2}(\Sigma R - R_{B1}) - R_{B3}(\Sigma R - R_{B3}) \\ n &= -R_{B2}(R_{B1} + R_o) & D_1 > D_2 \\ n &= -R_{B1}(R_{B2} + R_{B3}) & D_2 > D_1 \\ p &= -(R_{B2} + R_{B3})V_1 + (\Sigma R - R_{B2} - R_{B3})V_2 + (\Sigma R - R_{B2} - R_{B3})V_3 \end{aligned} \right.$$

$$\dot{\hat{x}} = \bar{A} \hat{x} + \left[\begin{aligned} &\frac{n(a_{21}u_1 - a_{11}u_2)}{L_1 \Sigma R(a_{11}a_{22} - a_{12}a_{21})} \\ &\frac{n(a_{12}u_2 - a_{22}u_1) + m(a_{21}u_1 - a_{11}u_2) + p(a_{11}a_{22} - a_{12}a_{21})}{L_2 \Sigma R(a_{11}a_{22} - a_{12}a_{21})} \end{aligned} \right] \hat{d}_2$$

$$s\hat{x}(s) = \bar{A}\hat{x}(s) + \begin{bmatrix} g_1 \\ g_2 \end{bmatrix} \hat{d}_2(s)$$

$$\begin{cases} (s-a_{11})\hat{i}_{L_1}(s) - a_{12}\hat{i}_{L_2}(s) = g_1\hat{d}_2(s) \\ -a_{21}\hat{i}_{L_1}(s) + (s-a_{22})\hat{i}_{L_2}(s) = g_2\hat{d}_2(s) \end{cases}$$

$$\hat{i}_{L_1}(s) = \begin{bmatrix} 1 & 0 \end{bmatrix} \hat{x}(s) = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} \hat{i}_{L_1}(s) \\ \hat{i}_{L_2}(s) \end{bmatrix}$$

$$\left(\begin{bmatrix} s & 0 \\ 0 & s \end{bmatrix} - \bar{A} \right) \hat{x}(s) = \begin{bmatrix} g_1 \\ g_2 \end{bmatrix} \hat{d}_2(s)$$

$$\hat{x}(s) = \left(\begin{bmatrix} s & 0 \\ 0 & s \end{bmatrix} - \bar{A} \right)^{-1} \begin{bmatrix} g_1 \\ g_2 \end{bmatrix} \hat{d}_2(s)$$

$$\hat{x}(s) = \frac{1}{(s-a_{11})(s-a_{22}) - a_{12}a_{21}} \begin{bmatrix} s-a_{22} & a_{12} \\ a_{21} & s-a_{11} \end{bmatrix} \begin{bmatrix} g_1 \\ g_2 \end{bmatrix} \hat{d}_2(s)$$

$$\hat{i}_{L_1}(s) = \frac{(s-a_{22})g_1 + a_{12}g_2}{(s-a_{11})(s-a_{22}) - a_{12}a_{21}} \hat{d}_2(s)$$

$$G = \frac{\hat{i}_{L_1}(s)}{\hat{d}_2(s)} = \frac{(s-a_{22})g_1 + a_{12}g_2}{s^2 - (a_{11} + a_{22})s + (a_{11}a_{22} - a_{12}a_{21})}$$